

Package ‘BayGMST’

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Type: Package
Title: Bayesian Reconstruction of Global Mean Surface Temperature
Version: 0.1.0
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License: GPL (≥ 3)

READ THIS FIRST

This is a **hand-typeset preview**, not the real, authoritative CRAN reference manual. It was produced in a sandbox with **no R installation available**, so it could not be generated the normal way (`devtools::document()` to build `man/*.Rd`, then `devtools::build_manual()` / R CMD Rd2pdf to render this PDF from those `.Rd` files). Content below was written by hand to mirror that output as closely as possible, from the same source descriptions as the `roxygen2` comments in R/.

Before treating this as final: run the two commands above (see `CRAN-READINESS.md` at the package root for the full checklist) and let the real output supersede this file.

Text marked **[GAP]** like [this](#) is a genuine information gap I could not resolve myself. Text marked **[FLAG]** like [this](#) is a decision, judgment call, or unresolved issue from the original code that needs Tyler’s or Julien’s sign-off, not something I decided unilaterally.

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1 Statistical Model

[FLAG] This section is not part of a standard CRAN Rd2pdf reference manual – those only ever contain per-topic help pages, never narrative model exposition. It’s included here because it was explicitly requested, to connect the code to the underlying statistics. Consider moving it into `vignettes/baygmst-intro.Rmd` (or a separate paper/preprint) rather than keeping it in the reference manual once the real, roxygen2-generated version of this document is built, since R CMD Rd2pdf has no mechanism to reproduce a hand-added section like this one.

The model implemented in `src/stan/baygmst.stan` (via `fit_baygmst()`) is the two-equation reduced-proxy Bayesian hierarchical model of Barboza et al. (2014, 2019) and Wang (2020), as described in Bagwell’s thesis proposal (Rice University, Dept. of Statistics, “Project II: Bayesian State-Space Reconstructions of GMST with Physically Motivated Ocean Memory,” §2.1). Using the proposal’s own notation (year subscript t), with P_t the reduced/representative proxy series and T_t the (possibly latent) global mean surface temperature:

Data level (proxy equation).

$$P_t = \phi_P P_{t-1} + \alpha_0 + \alpha_1 T_t + \sigma_P \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1) \quad (1)$$

Process level (temperature equation).

$$T_t = \phi_T T_{t-1} + \beta_0 + \beta_G G_t + \beta_V V_t + \beta_S S_t + \sigma_T \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, 1) \quad (2)$$

where G_t , V_t , S_t are the (already-transformed – see `transform_forcings()`) greenhouse-gas, volcanic, and solar forcings, and $|\phi_P|, |\phi_T| < 1$, $\sigma_P, \sigma_T > 0$.

Mapping to the code

Proposal notation	Stan parameter	R argument / role
$\overline{T}_t$	y (transformed parameter)	<code>instrumental_T</code> (observed part), <code>y_mis</code> (reconstructed part)
P_t	z (data)	<code>proxy</code> – see <code>reduce_proxies()</code>
ϕ_P	<code>phi_R</code>	—
σ_P	<code>sigma_z</code>	—
α_0, α_1	<code>alpha0</code> , <code>alpha1</code>	—
ϕ_T	<code>phi_T</code>	—
σ_T	<code>sigma_y</code>	—
$\beta_0, \beta_G, \beta_V, \beta_S$	<code>beta0</code> , <code>betaG</code> , <code>betaV</code> , <code>betaS</code>	—
G_t, V_t, S_t	G, V, S (data)	<code>forcing_G</code> , <code>forcing_V</code> , <code>forcing_S</code>

[FLAG] The Stan variable is named `phi_R/sigma_z` (“R” for “reduced proxy,” matching this specific implementation’s single-series design) rather than `phi_P/sigma_P` as in the proposal’s more general notation – same quantity, different name. I did not rename the Stan code to match the proposal, to avoid touching model code beyond what was already agreed; flagging the mismatch here instead.

Priors

As specified in the Stan model’s `model` block:

$$\begin{aligned} \alpha_0, \alpha_1, \beta_0, \beta_G, \beta_V, \beta_S &\sim N(0, 0.5) \\ \phi_P \text{ (phi_R)}, \phi_T &\sim N(0, 0.3) \\ \sigma_P \text{ (sigma_z)}, \sigma_T \text{ (sigma_y)} &\sim \text{Exponential}(1) \end{aligned}$$

[FLAG] These weakly informative priors are exactly as inherited from the original `BayGMST_v1.0.stan`; I have not evaluated their sensitivity or appropriateness – that judgment belongs to Tyler/Julien.

Simplification relative to the general proposal: no instrumental measurement-error layer

[GAP] The thesis proposal’s more general state-space formulation (§2.2, Eq. 5) adds a separate instrumental measurement equation $I_t = T_t + \sigma_I \xi_t$, treating the observed instrumental record I_t as a *noisy* realization of the latent true temperature T_t . **The model actually implemented here does not include that layer:** in `baygmst.stan`, `y[idx_obs] = y_obs` sets the latent state exactly equal to the raw instrumental values for observed years – i.e. instrumental temperature is treated as error-free. There is no σ_I parameter in this implementation. This is a real, substantive simplification relative to the proposal’s stated framework, not just a naming difference – worth flagging explicitly since someone reading the proposal and then the code could otherwise assume they match exactly.

Not yet implemented: the proposal’s main innovations

[GAP] The thesis proposal’s actual *proposed innovations* (§2.2–2.3) – (1) replacing the single reduced proxy P_t with the full multivariate proxy network $P_{i,t}$, $i = 1, \dots, m$, estimated via a Kalman filter/smoother with hierarchical partial pooling on proxy sensitivities $\alpha_{i,1}$, and (2) replacing the AR(1) temperature equation (Eq. 2) with a physically motivated multi-layer ocean-memory box model, $T_t = T_{t,\text{det}}(G, V, S) + \sigma \mathbf{c}^\top \mathbf{x}_t$ with $\mathbf{x}_t$ a vector of Ornstein-Uhlenbeck processes at timescales $\boldsymbol{\tau}$ – are **not implemented in this package**. They are the actual subject of Tyler’s proposed thesis work, still to be written. Nothing in BayGMST 0.1.0 should be read as already delivering them; `reduce_proxies()` still collapses the proxy network to a single series exactly as the *current* (not proposed) reduced-proxy approach does.

2 R topics documented:

BayGMST-package	BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature
cv_baygmst	K-fold cross-validate the BayGMST model over the instrumental period
fit_baygmst	Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model
plot_cv	Plot k-fold cross-validation reconstructions
plot_posterior_densities	Posterior density plots for a fitted BayGMST model
plot_reconstruction	Plot a BayGMST reconstruction
plot_trace	Trace plots for a fitted BayGMST model
print.baygmst_cv	Print method for "baygmst_cv" objects
print.baygmst_fit	Print a fitted BayGMST model
print.summary.baygmst_fit	Print method for "summary.baygmst_fit" objects
reconstruct	Extract a tidy reconstruction from a fitted BayGMST model
reduce_proxies	Reduce a multi-proxy network to a single representative proxy series
summary.baygmst_fit	Summarize a fitted BayGMST model
transform_forcings	Transform raw radiative forcings for use with <code>fit_baygmst()</code>

3 BayGMST-package

BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature

Description

Fits a Bayesian hierarchical AR(1) state-space model relating global mean surface temperature (GMST) to a reduced paleoclimate proxy network, instrumental temperature observations, and radiative forcings (greenhouse gases, volcanic aerosols, and solar irradiance). Missing pre-instrumental temperatures are estimated jointly with the model parameters using Stan via the `instantiate` and `cmdstanr` packages.

Model fitting requires CmdStan

`fit_baygmst()` and `cv_baygmst()` compile and sample a Stan model, which requires a working CmdStan installation. Check with `instantiate::stan_cmdstan_exists()` first. `reduce_proxies()` and `transform_forcings()` do not require CmdStan.

Provenance

Restructured from a collection of top-level analysis scripts (see `NEWS.md` for exactly what changed, and `inst/legacy-scripts/` for the untouched originals). **[GAP]** The **Description:** field's exact public-facing wording in `DESCRIPTION` was drafted by me from the `README` and the thesis proposal's Project II motivation section — it has not been reviewed or approved by Tyler or Julien. Please read and edit it before any submission.

Author(s)

Tyler Bagwell <teb6@rice.edu> (maintainer), Julien Emile-Geay <julieneg@usc.edu>. **[FLAG]** These are the only two contributors visible in `git log`; add others if appropriate.

4 transform_forcings

Transform raw radiative forcings for use with `fit_baygmst()`

Description

Applies the forcing transformations used throughout the BayGMST pipeline: volcanic forcing is mapped to a negative saturating form, greenhouse-gas (CO₂) forcing is log-transformed relative to a reference concentration, and solar forcing is centered on its own mean. Ported from the normalization block near the top of the original `BayGMST_v1.0.R` script.

Usage

```
transform_forcings(G, V, S, co2_c0 = 280, co2_coef = 5.35, vol_coef = 25)
```

Arguments

G	Numeric vector of raw greenhouse-gas forcing (e.g. atmospheric CO ₂ concentration in ppm).
V	Numeric vector of raw volcanic forcing (e.g. aerosol optical depth). Same length as G .
S	Numeric vector of raw solar forcing (e.g. total solar irradiance). Same length as G .
co2_c0	Reference/baseline CO ₂ concentration used in the log transform. Default 280 (ppm).
co2_coef	Multiplier applied to the log CO ₂ ratio. Default 5.35.
vol_coef	Multiplier applied to the volcanic saturating transform. Default 25.

Details

$$V' = -|\text{vol_coef}| \times (1 - e^{-V}), \quad G' = \text{co2_coef} \times \log(G/\text{co2_c0}), \quad S' = S - \bar{S}$$

Value

A list with three numeric vectors, **G**, **V**, and **S** – the transformed forcings.

Examples

```
fc <- transform_forcings(
  G = c(280, 300, 340, 400),
  V = c(0, 0.05, 0.02, 0.10),
  S = c(1360, 1361, 1360.5, 1361.2)
)
fc$G
```

5 reduce_proxies

Reduce a multi-proxy network to a single representative proxy series

Description

Collapses a matrix of paleoclimate proxy records into a "representative proxy" series suitable for `fit_baygmst()`, following the segmented-composite approach of Barboza et al. (2019), ported from `utils/PAGES2k_reducedProxy_UNSC.R`. The proxy network is broken into overlapping time segments of `chunk` years; within each segment, available proxies are regressed against instrumental temperature during `calib_years` using the chosen `method`. The final composite series averages across all segments covering a given year.

Usage

```
reduce_proxies(proxy_matrix, years, temp_calib, calib_years,
method = c("PCR", "LASSO", "sPCR", "SPLS", "SIR"),
chunk = 250, target_adj_r2 = 0.7, max_na_frac = 0.05,
n_components = 3, spls_eta = seq(0.1, 0.9, 0.1))
```

Arguments

<code>proxy_matrix</code>	Numeric matrix/data.frame, one row per year (matching <code>years</code>), one column per proxy. No year column.
<code>years</code>	Integer vector of calendar years, one per row of <code>proxy_matrix</code> .
<code>temp_calib</code>	Numeric vector, instrumental temperature during <code>calib_years</code> .
<code>calib_years</code>	Integer vector, calibration years (subset of <code>years</code> ; <code>length(calib_years) == length(temp_calib)</code>).
<code>method</code>	One of "PCR" (default), "LASSO", "sPCR", "SPLS", or "SIR". "SIR" is experimental — see Details.
<code>chunk</code>	Segment length in years. Default 250. Every segment must fully contain <code>calib_years</code> , or <code>reduce_proxies()</code> errors (see Details).
<code>target_adj_r2</code>	For "PCR": adjusted- R^2 stopping threshold for adding principal components. Default 0.70.
<code>max_na_frac</code>	Proxies missing more than this fraction of a segment are dropped from it. Default 0.05.
<code>n_components</code>	For "sPCR" only: number of supervised components (scalar or per-segment vector). Default 3.
<code>spls_eta</code>	For "SPLS" only: sparsity grid passed to <code>spls::cv.spls()</code> .

Details

Method support. "PCR", "LASSO", and "SPLS" are fully adaptive and supported. "sPCR" is supported but requires `n_components` to be chosen by the caller (not cross-validated, matching the original script).

[FLAG] "SIR" is a simplified, *unvalidated* reimplementation using a single sliced-inverse-regression direction from `dr`, with no per-segment variable pre-selection. It intentionally does *not* reproduce the original script's SIR branch, which called an undocumented `edrSelec()` step from the non-CRAN package 'edr-GraphicalTools' that the original author flagged as never fully working ("NEED TO FIGURE OUT HOW TO INSTALL `edrGraphicalTools`!!"). `config.yml` in the original repository likewise documented SIR as "still under construction." Treat `method = "SIR"` output with corresponding caution.

[FLAG] The segmentation/calibration-window constraint (`chunk` must be small enough that even the shortest segment still spans all of `calib_years`) mirrors an explicit `stop()` check in the original script. I ported

that check explicitly rather than letting a mismatched-length input fail with a cryptic error deep inside `lm()` — but the underlying design constraint itself is inherited from the original code, not something I introduced.

Required packages: "LASSO" needs `glmnet`; "sPCR" needs `superpc`; "SPLS" needs `spls`; "SIR" needs `dr`. All are **Suggests**, only required for the method actually called.

Value

A list with **segments** (data.frame, one column per segment), **composite** (data.frame with columns **year** and **RP1**, the series to pass to `fit_baygmst()`), and **method**.

6 fit_baygmst

Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model

Description

Prepares the data list expected by the `baygmst` Stan model and samples from it with `cmdstanr`, via `instantiate`. Ported from the data-preparation and `mod$sample()` block of the original `BayGMST_v1.0.R` script, restructured to take explicit arguments instead of reading `config.yml` and hardcoded relative paths, and to return an object rather than leaving variables in the global environment.

Usage

```
fit_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

<code>proxy</code>	Representative proxy series – see <code>reduce_proxies()</code> . No missing values.
<code>instrumental_T</code>	Instrumental temperature, NA for pre-instrumental years to reconstruct.
<code>forcing_G</code> , <code>forcing_V</code> , <code>forcing_S</code>	Already-transformed forcings – see <code>transform_forcings()</code> . No missing values.
<code>years</code>	Calendar years, defining row order for all arguments above.
<code>chains</code> , <code>parallel_chains</code> , <code>iter_warmup</code> , <code>iter_sampling</code> , <code>seed</code>	Passed to <code>cmdstanr</code> 's <code>\$sample()</code> . <code>iter_sampling</code> $\geq$ 1000 required.
<code>...</code>	Additional arguments passed to <code>\$sample()</code> .

Value

An object of class "baygmst_fit": a list with `fit` (the `CmdStanMCMC` object), `years`, `idx_obs`, `idx_mis`, `data`, and `call`.

Note

[GAP] This function does not write any files by default (no `auto-ggsave()`, no auto-CSV export) – a deliberate deviation from the original script, which wrote directly to `cfg$folder_paths$...`. This matches CRAN policy on functions not writing to the filesystem by default, but it does mean any code that relied on those side-effect files (e.g. `outputs/reconstructions/fit_post_summaries.csv`) needs to be updated to call `write.csv()` itself on the returned object.

7 cv_baygmst

K-fold cross-validate the BayGMST model over the instrumental period

Description

Partitions the instrumental years of `instrumental_T` into `nfold` contiguous blocks; for each, refits the model with that block hidden, and reports recovery accuracy. Ported from the fold loop in `cv_v0.1.R`.

Usage

```
cv_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  nfold = 3, chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

Same as `fit_baygmst()`, plus `nfold` (number of folds, default 3).

Value

An object of class `"baygmst_cv"`: `folds` (combined data.frame of held-out predictions), `r2` and `mse` (per-fold, posterior mean of the Stan-computed `r2_cv/mse_cv` generated quantities), `years`, `call`.

Note

[FLAG] Two blocks in the original `cv_v0.1.R` were dropped, not ported: (1) a `Forcings.projections/RCP-scenario` block that was built but never referenced again (dead code), and which additionally hardcoded a path to a specific contributor's machine (`/Users/tylerbagwell/...`); and (2) a large commented "OLD" duplicate of the fold loop at the end of the file. Confirm nothing downstream still depended on either.

[FLAG] This implementation reports the fully Bayesian posterior mean of the Stan model's own `r2_cv/mse_cv` generated quantities. The original (non-dead) R loop instead computed a plug-in point-estimate R^2 in R from the posterior mean reconstruction. The two will be close but not identical – flagging the substitution explicitly rather than silently changing what "R2" means.

8 reconstruct

Extract a tidy reconstruction from a fitted BayGMST model

Description

Builds a tidy data.frame combining observed instrumental temperatures with the posterior reconstruction of pre-instrumental years – equivalent to the `gmst_reconstruction_data.csv` the original script wrote to disk as a side effect, but returned as an object instead.

Usage

```
reconstruct(fit, probs_inner = c(0.16, 0.84), probs_outer = c(0.025, 0.975))
```

Value

A data.frame with columns `year`, `type` ("instrumental"/"reconstruction"), `T_obs`, `T_mean`, `T_lo_inner`, `T_hi_inner`, `T_lo_outer`, `T_hi_outer`.

9 `plot_reconstruction`, `plot_trace`, `plot_posterior_densities`, `plot_cv`

Plotting functions

Description

- `plot_reconstruction(fit, title = "GMST Reconstruction")` – ported from the `p_ts` block of BayGMST_v1.0.R.
- `plot_trace(fit, parameters = ...)` – ported from the trace-plot block.
- `plot_posterior_densities(fit, parameters = ...)` – ported from the posterior-histogram block.
- `plot_cv(cv)` – ported from the `p_ts_cv` block of `cv_v0.1.R`.

All four return a `ggplot` object instead of calling `ggsave()`/auto-printing as a side effect; save with `ggplot2::ggsave()` if a file is wanted. **[FLAG]** The original script combined the reconstruction and posterior-density panels into one figure via `patchwork (p_ts + p_hist)`. Splitting these into separate functions means that combination is no longer automatic – users who want the original combined layout can do `plot_reconstruction(fit) | plot_posterior_densities(fit)` themselves via `patchwork` (moved to `Suggests`, since the package itself no longer calls it directly).

10 print/summary methods

S3 methods

Description

- `print.baygmst_fit(x, ...)` – one-line-per-field overview (year range, chains/draws, call).
- `summary.baygmst_fit(object, ...)` – posterior parameter summary (`alpha0`, `alpha1`, `phi_R`, `phi_T`, `beta0`, `betaG`, `betaS`, `betaV`, `sigma_y`, `sigma_z`) plus instrumental-period fit performance (RMSE, MAE, Bias, Correlation, R^2 , detrended R^2), replacing the `perf_stats/R2_dt` block of `BayGMST_v1.0.R`.
- `print.summary.baygmst_fit(x, ...)` – pretty-prints the above.
- `print.baygmst_cv(x, ...)` – per-fold and median R^2 .

Known limitations index

[FLAG] `src/stan/baygmst.stan`'s `y_ins_fitted` block (`t == 1` branch) indexes `y/z` with the count `NT_mis` rather than `idx_mis[NT_mis]`, flagged by the original author with `// CHECK THIS LINE`. On inspection (see §1 and `NEWS.md`), this is numerically correct given how this pipeline's inputs are always laid out (missing years contiguous at the start, observed years contiguous at the end) – *not* the live bug I initially reported it as – but it is fragile, undocumented shorthand that would break under non-contiguous missingness, and its effect is narrowly scoped to `y_ins_fitted[1]` alone, not to the parameter posteriors or to `y_mis`. Left unchanged, since even a belt-and-suspenders self-documenting rewrite is a code change to model machinery that deserves Tyler/Julien's sign-off, not a unilateral one.

[GAP] The thesis proposal's actual proposed innovations for Project II – the multivariate multi-proxy Kalman filter/smoothing and the physically motivated multi-box ocean-memory temperature process – are not implemented in this package version. See §1.

[FLAG] Several `library()` calls in the original scripts (`car`, `fda`, `ggmap`, `maps`) had no evident corresponding usage in the code as written and were not carried into `DESCRIPTION`. If a hidden use is found, re-add them.

[GAP] `DESCRIPTION`'s `RoxygenNote` field is a placeholder (7.3.2) – `devtools::document()` will set the real value.

[GAP] `NAMESPACE` and every file in `man/` are hand-authored placeholders (see the header comment in each), not actual `roxygen2` output, because this restructuring was done without an R installation available to run it. `CRAN-READINESS.md` has the exact commands to regenerate and verify them for real.